

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Civil Engineering

Questions: 15

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

1) (12 — $r^2 \setminus 4$ where v = velocity of water at any position r μ = constant = viscosity of water d = pipe diameter r = radial distance from centreline of pipe What is the shear stress at the wall of the pipe due to the water? What is the shear stress at a position $r = d/4$? If the given profile persists a distance L along the pipe, what drag is induced on the pipe by the water in the direction of flow over this distance ? (e) Water flows in a trapezoidal channel having a bottom width of 4.57 m and side slopes of 1 vertical to 1-5 horizontal at a rate of 5.66 m³/s. If the depth of flow is 0.61 m, determine whether the flow is subcritical or supercritical. Take $g = 9.81 \text{ m/s}^2$. FSI-P-CVLE 7 ===== PAGE 8 ===== 0 Effective overburden stress, kN/m² 100 200 300 400 500

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2. [UNKNOWN] paper-i

Q2. (a) A prestressed concrete beam supports an imposed load of 5 kN/m over an effective span of 12 m. The beam has a rectangular section with a width of 250 mm and a depth of 700 mm. Find the effective prestressing force in the cable if it is parabolic with an eccentricity of 110 mm at the centre and zero at the ends, for the following conditions : if the bending effect of prestressing force is nullified by the imposed load for the mid-span section by neglecting the self weight of the beam. if the resultant stress due to self-weight, imposed load and prestressing force is zero at the soffit of the beam for the mid-span section. Assume the density of concrete is 25 kN/m³. 8 8 10 FSI-P-CVLE 3 ===== PAGE 4 ===== (i) A shaft is subjected to a maximum torque of 10 kNm and a maximum bending moment of 7.5 kNm at a particular section. If the allowable equivalent stress in simple tension is 160 MN/m², find the diameter of the shaft according to the maximum shear stress theory. 10 (ii) State maximum principal stress theory. 5 (i) The effective length of compression flange of a simply supported beam ISMB -600 @ 1.225 kN/m is 10 m. Determine the safe uniformly distributed load per metre length which can be placed over the beam having an effective span of 10 metres. The ends of the beam are restrained against rotation at the bearing. The maximum permissible stresses are given below. $\sigma_y = 250 \text{ Mpa}$. /0 Properties of ISMB 600 @ F225 kN/m. Area = 15600 mm² $Z = 3060 \times 103 \text{ mm}^3$ Mean thickness of compression flange, $t_f = 20.8 \text{ mm}$ Thickness of web, $t_w = 12 \text{ mm}$ Maximum thickness of flange, $h_2 = 45.15 \text{ mm}$ Radius of gyration $r_x = 242 \text{ mm}$ $r_y = 4F2 \text{ mm}$ YY Maximum permissible bending stress $\sigma_{b\text{max}} = 160 \text{ N/mm}^2$ in equal flange I-beams D / t_f -4 16 18 20 25 30 / / r_{yy} 4, 120 126 122 119 113 109 130 121 118 114 108 103 140 118 113 110 103 97 150 114 109 105 98 92 160 110 106 101 93 87 170 107 102 98 89 ' 83 180 104 99 94 85 79 190 101 95 91 82 75 FSI-P-CVLE 4 ===== PAGE 5 ===== (ii) An RC wall of 200 mm thickness and 3.6 m effective height is needed for a compressive load of 1250 kN. Design the wall using M 15 grade of concrete and Fe 415 steel reinforcement.

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3. [UNKNOWN] paper-i

Q3. (a) Draw the influence line diagram (ILD) for bending moment and shear force for a section at 5 m from the left hand support of a simply supported -beam, 20 m long. Hence calculate the maximum bending moment and shear force at the section, due to uniformly distributed rolling load of length 8 m and intensity 10 kNm run. Design the vertical wall of a cantilever retaining wall to retain earth embankment 4.5 m above ground level. The density of earth is 16 kN/m³ and its angle of repose is 30°. Top of the embankment is levelled one. The safe bearing capacity of the soil may be taken as 210 kN/m² and the coefficient of friction between soil and concrete is 0.58. Adopt M 20 grade concrete and Fe 415 grade steel. A flywheel of diameter 0.7 m has a mass of 80 kg as shown in the figure below. The coefficient of friction between the band and the flywheel is 0.40. If the initial rotating velocity of the flywheel is 400 rpm clockwise, determine the magnitude of the force P required to stop the flywheel in 5 seconds. 5 10 15 15 FSI-P-CVLE 5 ===== PAGE 6 =====

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4. [UNKNOWN] paper-i

Q4. (a) Calculate the load carrying capacity of a rolled steel section ISMB 600. The height of the column is 8 m; fixed at base and pinned at top. The yield stress of steel is 250 MPa. The permissible compressive stresses are as given in the table below: 10 Slenderness ratio (X) 90 100 110 120 130 140 150 160 170 180 Allowable compressive stress in MN. (ccad 90 80 72 64 57 51 45 41 37 33 The cross-section properties of ISMB 600 section are as follows : Area of the cross section = 15600 mm² Flange width = 210 mm Flange thickness = 20.8 mm Web thickness = 12 mm = 91813 x 106 mm⁴ I = 26.5 x 106 mm⁴ - YY Analyse the continuous beam shown in the figure below by flexibility method and draw the bending moment diagram. Take EI = Constant. 15 1100 kN AC 1g-- 4m +1 5 m+1-5 m->1 Design an RC floor slab for a room of internal dimensions of 3.5 m x 8.5 m. Assume the slab to be simply supported on 230 mm thick masonry walls. The slab is to support a live load of 4 kN/m² and a floor finish load of 1.5 kN/m². Use M 25 grade of concrete and Fe 415 grade steel. Draw the details of reinforcements. The band strength for M 25 concrete (Tbd) is 1.26 N/mm². The design shear strength of concrete τ_c is given as follows for different values of P = 100A_{sc}/b_d 15 100 A_{sc} < 0.15 0.25 0.50 0.75 b_d -cc 2 N/mm 0.29 0.36 0.49 0.57 FSI-P-CVLE 6 ===== PAGE 7 ===== SECTION B

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5. [UNKNOWN] paper-i

Q5. (a) What are important clay minerals ? Why do clays exhibit plasticity? 8 (b) A granular soil was tested in the laboratory and found to have maximum and minimum void ratios of 0.82 and 0.38 respectively. The specific gravity of soil solids was 2.66. A natural deposit of the same soil has 10% moisture and its moist unit weight is 19 kN/m³. Determine the relative density of the soil in the field. 8 (c) What are the merits and demerits of the stone column method of soil improvement? 8 (d) Water is moving through a pipe. The velocity profile at a particular section is given mathematically as $V =$

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6. [UNKNOWN] paper-i

Q6. (a) An SPT test was conducted in a sand deposit at 12 m depth. The unit weight of sand is 20 kN/m³ and water table is at 2 m below natural ground surface. If the observed SPT value is 24; what will be the corrected N value ? Use the figure below to apply overburden correction. 10 Correction factor CN 0 2 0.4 0.6 0 8 1.0 1.2 1.4 1.6 1.8 2.0 Correction factor for N-values for overburden correction (b) Forest areas usually have freshwater ponds and lakes in which small animals such as crayfish, shrimp, etc. live. One such animal has an average diameter of 1 mm. We want to know the drag force on this animal when it moves slowly in freshwater. As it is difficult to measure this drag force on such a small animal, a scale model 100 times larger is made and tested in glycerin in a laboratory at V = 30 cm/s. The measured drag on the model is 1.3 N. For similar conditions, find the velocity and drag of the actual animal in water. Given that 15 $\rho_{\text{water}} = 0.001 \text{ kg/(m-s)}$ $\rho_{\text{water}} = 999 \text{ kg/m}^3$ $\rho_{\text{glycerin}} = 1263 \text{ kg/m}^3$ FSI-P-CVLE 8 ===== PAGE 9 ===== (c) (i) Sketch the general layout of a hydropower system with a reaction turbine. (ii) A reaction turbine is supplied with water through a 150 cm pipe (penstock) (e = 1.0 mm) that is 40 m long. The water surface in the reservoir is 20 m above the draft tube inlet, which is 4-5 m above the water level in the tailrace. If the turbine efficiency is 92% and the discharge is 12 m³/s, what is the power output of the turbine in kilowatts ? Use $f = 0.0185$, $g = 9.81 \text{ m/s}^2$. 15

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7. [UNKNOWN] paper-i

Q7. (a) A square footing of 2 m x 2 m is placed in a soil deposit at 1.5 m depth. The soil properties are $c = 60 \text{ kN/m}^2$, $\phi = 22^\circ$, $\gamma = 20 \text{ kN/m}^3$. What maximum vertical load can this footing carry with FOS = 3 ? If the footing is subjected to moment of 160 kN-m in one direction and 130 kN-m in the other direction, what will be the revised factor of safety ? Adopt bearing capacity factors from the table given below: 15 4? N_e N_q N_y 10 8.35 2.47 1.22 15 10-98 3.94 2.65 20 14.83 6-40 5-39 25 20.72 10.66 10.88 (b) A rectangular area of 4 m x 2 m is uniformly loaded with a pressure of 100 kPa at the ground surface. Calculate the vertical pressure at a point 3 m below one of its corners by equivalent-area method. Compare the stress at the same point by Boussinesq's 2 : 1 approximate stress distribution method. 15 FSI-P-CVLE 9 ===== PAGE 10 ===== (c) It is frequently desirable to install automatic gates to prevent a flood. One such gate ABC is shown in the figure. A 0.6 m 0.4m (Not to scale) The gate ABC is 1 m square and hinged at B. It will open automatically when water depth h becomes high enough. Determine the minimum value of h at which the gate will open. 10

8. [UNKNOWN] paper-i

Q8. (a) A 4 m high wall is shown in the figure. Determine the Rankine passive force per unit length of the wall. 15 $y = 16$ kN/m³ $c' = 0$, $\phi' = 30^\circ$ GWT $\gamma_{sat} = 18$ kN/1113 $c' = 10$ kN/m² $\phi' = 26^\circ$ (b) A cast iron pipe with 20 cm diameter and 15 mm wall thickness is carrying water when the outlet is suddenly closed. Given that $E_{water} = 217 \times 10^9$ N/m², $E_p = 16 \times 10^{11}$ N/m², $\rho = 1000$ kg/m³ and design discharge is 40 Us, calculate the water hammer pressure rise for 15 (i) rigid pipe walls, assuming sudden valve closure, and consider stretching of pipe walls, neglecting the longitudinal stress. FSI-P-CVLE 10 ===== PAGE 11 ===== vena contracta (c) Consider a vertical sluice gate in a wide rectangular channel. The depth of flow at the vena contracta is 0.457 m for a flow rate of 4.646 m³/s per metre of width. The channel bed slope is 0.0003 and Manning's roughness factor is $n = 0.020$. The depth of flow at point b is 0.5 m. Determine whether a jump occurs. What is the type of water surface profile? /0 Take $g = 9.81$ m/s². FSIT-CVLE 11

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9. [UNKNOWN] paper-ii

Q2. (a) Differentiate between the following: 15 Plastering and Pointing Balcony and Terrace Roof decking and Roof covering FSI-D-CVLE 2 ===== PAGE 3 ===== Two straights PQ and QR intersect at a chainage of 3150 m. The angle of intersection is 130°. It is required to set out a 4° simple circular curve to connect the straights. The chain used for setting out the curve is of 30 m length. The length of first subchord is 25.25 m and length of last subchord is 200m. Calculate all the necessary data required for setting out the curve using the method of offsets. A fixed time two-phase signal is to be designed for a four-legged intersection, for which the design hourly flow and saturation flow are given below: Details of flow North South East West Design hourly flow (veh/h) 570 320 550 455 Saturation flow (veh/h) 1750 1500 2210 2265 Time lost per phase due to starting delays is 2 seconds. Calculate the optimum cycle time and allocate the green time for the two phases using Webster's method.

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10. [UNKNOWN] paper-ii

Q3. (a) What are the merits and demerits of line and staff organisation for a construction company? 10 What are the main constituents of a concrete batching and mixing plant? 10 An upgrade road section of gradient 1 in 130 meets a downgrade road section of gradient 1 in 130. A summit curve is required to be designed for a design speed of 65 km/h so that an overtaking sight distance of 350 m is available. Determine the length of the summit curve. 10 Design a tie-bar system for a cement concrete pavement with the following data: 10 Thickness of slab = 18 cm Width of slab = 3.30 m Coefficient of friction = 1.5 Unit weight of pavement = 2400 kg/m³ Allowable working stress in steel = 1400 kg/cm² Maximum permissible bond stress in concrete = 17.5 kg/cm² Maximum permissible tensile stress in steel bars = 24.6 kg/cm² FSI-D-CVLE 3 15 10 ===== PAGE 4 =====

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11. [UNKNOWN] paper-ii

Q4. (a) A shopping centre is proposed to be developed on a valuable farmland. The farmland presently contributes 10 crores per year. It is expected that the proposed shopping centre will have a benefit of 70 crores per year. It is proposed that an amount of 500 crores is to be deposited in the bank for a period of 50 years where it will earn interest at a rate of 10% per year compounded annually. A fixed amount will be withdrawn yearly as a yearly development cost so that the amount of 500 crores will just be depleted at the end of 50 years. Analyse the desirability of the above project proposal, assuming a yearly maintenance cost of 2 crores also. Further, assuming that the affected farmers may demand to compensate them for the cost of farmland. For this purpose, a fixed sum of money will be deposited in the bank at the end at each year such that a total amount of 120 crores will be accumulated at the end of the 10-year period to be paid to farmers. Assume interest rate as 5% per year, compounded annually. Would the revised project proposal be desirable, with this new disbenefit? 10 What are the harmful effects of cracks in buildings? What are the measures that can be taken to repair cracks in buildings? /0 Assuming a linear speed density relationship, the mean free flow speed and jam density are observed as 75 km/h and 90 vehicles/km. Compute the average time headway, spacing, clearance and gap when the flow is maximum. Take average length of vehicle as 6.1 m. 10 A horizontal circular curve of radius 320 m is to be aligned on a two-lane highway, 7 m wide in an urban area having the mixed traffic condition. The design speed is 65 km/h and the wheel base of the design vehicle is of 6 m length. Superelevation is achieved by rotating the

pavement about its centre line. Assume rate of introduction of superelevation as 1 in 100 for plain terrain. Determine the length of the transition curve. /0 FSI-D-CVLE 4 ===== PAGE 5 ===== 50 V 53 20 00 7.0 45 SECTION B

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12. [UNKNOWN] paper-ii

Q5. (a) What do you mean by flood routing? Distinguish between reservoir and channel routing. What information is required for routing a flood through a reservoir. List the name of two methods generally used for reservoir routing. (b) List various types of dams based on hydraulic design and mode of resistance offered by them against external forces. Calculate stresses developed at heel and toe of the dam profile shown in Figure 1. Consider only weight of dam and water pressure. Take unit weight of dam material = 24 kN/m^3 unit weight of water = 10 kN/m^3 (All levels and dimensions are in metres) Figure 1 Differentiate between River training and Cross drainage works. Classify cross drainage works based on the condition of crossing. River water is used as the source of water supply in a town. The major catchment of the river is agricultural fields and there is no industry existing in the catchment. What are the water quality parameters one needs to worry about? Why? To meet the drinking water quality, suggest the necessary treatment systems and explain why each system is needed. FSI-D-CVI F 5 ===== PAGE 6 ===== (e) Sedimentation tanks are used for various water and wastewater treatment facilities. Explain the design criteria for a secondary sedimentation tank of a wastewater treatment system. How is this different from the design of a primary sedimentation tank used for water treatment? 8

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13. [UNKNOWN] paper-ii

Q6. (a) (i) What is the significance of break point chlorination? Draw the titration curve representing Residual Chlorine vs Chlorine added in the following cases and explain the behaviour: Distilled water, 1000 mg/L sterilized glucose solution and Secondary treated wastewater. 5 (ii) Clean water at 20°C is passed through a bed of uniform sand at a filtering velocity of 6.0 m/h . The sand grains are 0-4 mm in diameter with a shape factor of 0.9 and a specific gravity of 2.6. The depth of the bed is 0.75 m and the porosity is 0.4. Determine the head loss through the bed. If the filter medium is to be expanded to a porosity of 0.6 by hydraulic back wash, determine the required back wash velocity and depth of expanded bed. Density of water at 20°C = 998.2 kg/m^3 . Dynamic viscosity of water at 20°C , $\mu = 0.01002 \times 10^{-3} \text{ N.s/m}^2$. Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$. 10 (b) (i) What do you mean by atmospheric stability? How is it affecting the ambient air quality? Explain. 5 (ii) A town has a population of 10,00,000 (10 lakhs). The municipality decided to construct a wastewater treatment plant using Activated Sludge Process (ASP). Show the flow diagram of the treatment system. Also calculate the volume of the aeration tank and the daily sludge wastage needed from the plant, if the BOD_u of the wastewater entering the ASP is 250 mg/L and the desired effluent quality is 10 mg/L as BOD_u. Assume that the system is a completely mixed reactor yield coefficient, $Y = 0.5$; decay constant of microbes, k_d is 0.05/d, MLSS concentration in aeration tank is 3000 mg/L and an underflow concentration of 10,000 mg/L is desired. The Biological Sludge Residence Time (BSRT) of the system, θ , is 10 days. 10 FSI-D-CVLE 6 ===== PAGE 7 ===== (c) Explain Darcy's law of flow through porous media. A confined stratified aquifer has a total thickness of 10 m and is made up of three layers. The bottom layer has coefficient of permeability of 25 m/day and a thickness of 4.0 m. The middle and top layers have permeability of 15 m/day and 30 m/day respectively and are of equal thickness. Calculate the equivalent permeability and transmissivity of the confined aquifer, if flow is along the stratification. 10

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14. [UNKNOWN] paper-ii

Q7. (a) (i) What are the salient features of Lacey's regime theory. Using Lacey's theory, design an irrigation channel for the following data: 10 Discharge (Q) = 40 cumec Silt factor (f) = 1 Side slopes = $1 : 1.2$ (ii) Briefly discuss the factors affecting infiltration capacity of soil. 5 (i) What is flow mass curve? How is it prepared? Explain the procedure of determining the storage capacity of a reservoir by using flow mass curve. 10 (ii) Briefly explain various purposes of providing galleries in dams 5 (i) Explain various phases of composting and the factors affecting composting process. 5 (ii) Draw a cross-sectional view of an engineered landfill and show various components. Also, explain the purpose of each component. 5

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15. [UNKNOWN] paper-ii

Q8. (a) (i) Given below are the ordinates of 2-hour (h) Unit Hydrograph (UH) for a catchment. Time (h) 0 2 4 6 8 10 12 14 16 Ordinates of 2-h UH, m³/s 0 8 25 30 22 16 8 5 0 A storm of 4-hour duration and having Excess Rainfall (ER) of 3.0 cm occurred on the same catchment. Calculate the ordinates of resulting flood hydrograph. Assume a constant base flow of 5 m³/s. 10 (ii) List various components of a barrage. Explain the role of pocket or approach channel in controlling the sediment flow into the canal. 5 FSI-D-CVLE 7 ===== PAGE 8 ===== (i) Differentiate between the following: 6 l. Design of Water distribution system and Sewer system Specific yield and Storage coefficient Hydraulic conductivity and Transmissivity (ii) What are the air pollutants emitted from a thermal power plant? To control these pollutants, what are the pollution control devices one can use ? Show the order in which these units need to be placed and why so? 9 (i) What do you mean by sustainable development ? How can the sustainability approach be used for waste management? 4 (ii) Determine the 3-day BOD and ultimate first stage BOD for a wastewater whose 5-day 20°C BOD is 250 mg/L. The reaction constant k (base e) is 0.23/d. What would have been the 5-day and 3-day BOD, if the test had been conducted at 28°C? 6 FSI-D-CVLE 8

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